

Molecular Dynamics: Workflow for MD Simulations

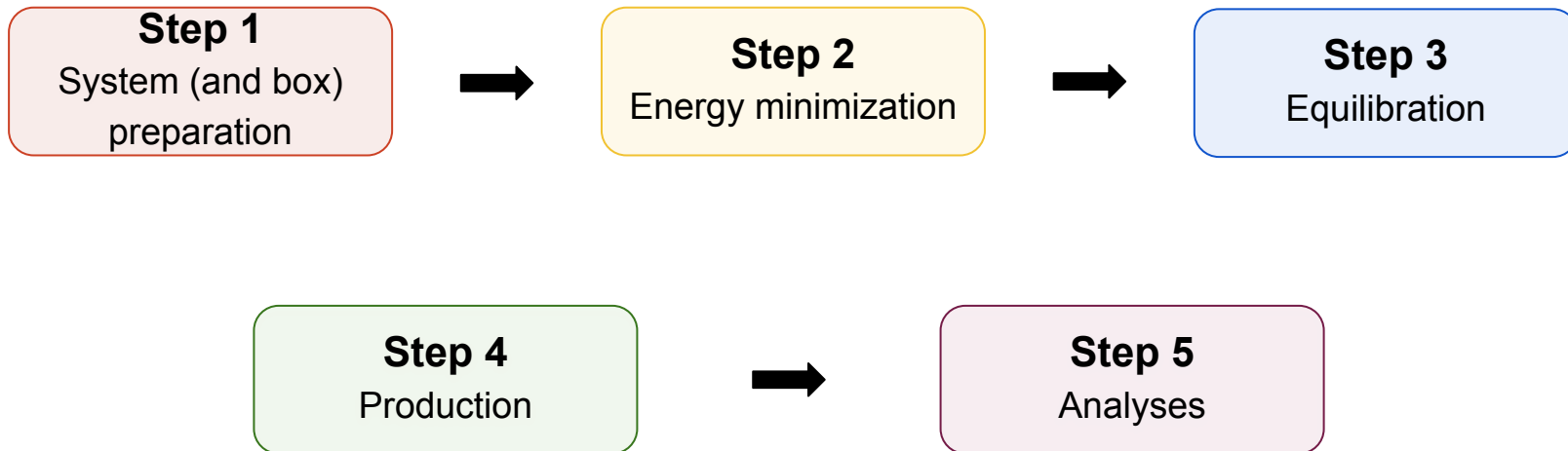
PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

Day 2/5

Dr. Marta Monti

Full MD pipeline



There are many softwares to run classical MD simulations (e.g., AMBER, LAMMPS, CP2K, etc...).

Among them, one of the most used is **GROMACS**



Step 1: System and box preparation

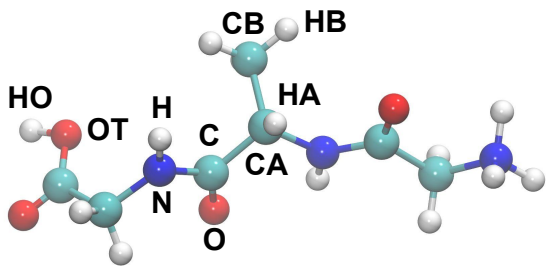
The first thing we need to do is prepare a solvated simulation box containing our biomolecule

Structure



Topology

Assign atom types, bonds, charges and force-field parameters to the biomolecule.



```
[ atoms ]
; nr      type  resnr residue  atom  cgnr  charge  mass  typeB  chargeB  massB
; residue  0 GLY rtp GLY  q +1.0
  1  opls_287  0    GLY    N    1    -0.3   14.0027
  2  opls_290  0    GLY    H1   2     0.33   1.008
  3  opls_290  0    GLY    H2   3     0.33   1.008
  4  opls_290  0    GLY    H3   4     0.33   1.008
  5  opls_292B  0    GLY    CA   5     0.19  12.011
  6  opls_140  0    GLY    HA1   6     0.06   1.008
  7  opls_140  0    GLY    HA2   7     0.06   1.008
  8  opls_235  0    GLY    C    8     0.5    12.011
  9  opls_236  0    GLY    O    9    -0.5   15.9994 ; qtot 1
; residue  0 ALA rtp ALA  q 0.0
 10  opls_238  0    ALA    N   10    -0.5   14.0067
 11  opls_241  0    ALA    H   11     0.3   1.008
 12  opls_224B  0    ALA    CA  12     0.14  12.011
 13  opls_140  0    ALA    HA  13     0.06   1.008
 14  opls_135  0    ALA    CB  14    -0.18  12.011
 15  opls_140  0    ALA    HB1  15     0.06   1.008
 16  opls_140  0    ALA    HB2  16     0.06   1.008
 17  opls_140  0    ALA    HB3  17     0.06   1.008
 18  opls_235  0    ALA    C   18     0.5    12.011
 19  opls_236  0    ALA    O   19    -0.5   15.9994 ; qtot 1
; residue  0 GLY rtp GLY  q 0.0
 20  opls_238  0    GLY    N   20    -0.5   14.0067
 21  opls_241  0    GLY    H   21     0.3   1.008
 22  opls_223B  0    GLY    CA  22     0.08  12.011
 23  opls_140  0    GLY    HA1  23     0.06   1.008
 24  opls_140  0    GLY    HA2  24     0.06   1.008
 25  opls_267  0    GLY    C   25     0.52  12.011
 26  opls_269  0    GLY    OT  26    -0.44  15.9994
 27  opls_268  0    GLY    O   27    -0.53  15.9994
 28  opls_270  0    GLY    HO  28     0.45   1.008 ; qtot 1
```

Step 1: System and box preparation

The first thing we need to do is prepare a solvated simulation box containing our biomolecule

Structure



Topology

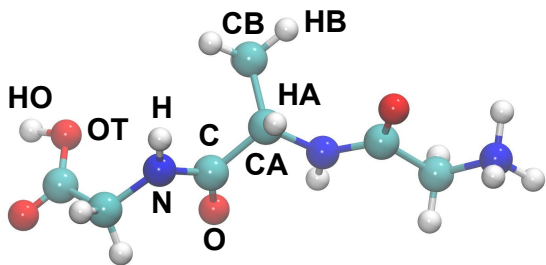


Simulation
box



Solvation
(H₂O + Ions)

Assign atom types, bonds, charges and force-field parameters to the biomolecule.



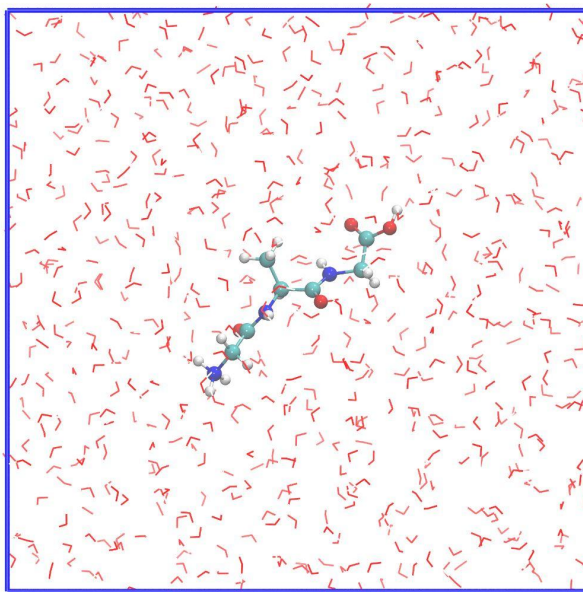
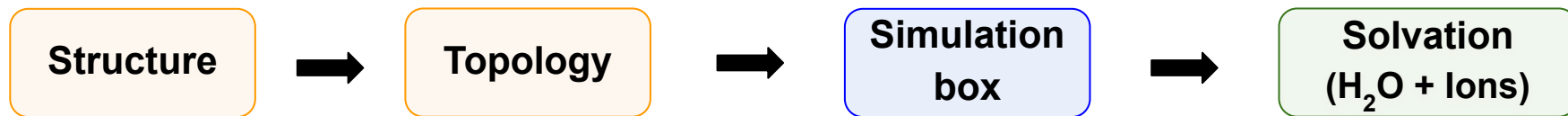
Place the biomolecule at the **center** of a box large enough to avoid interactions with its periodic images.

Fill the box with water and ions to obtain an electrically **neutral box**.

All these steps can be performed with GROMACS commands that we'll use this afternoon

Step 1: System and box preparation

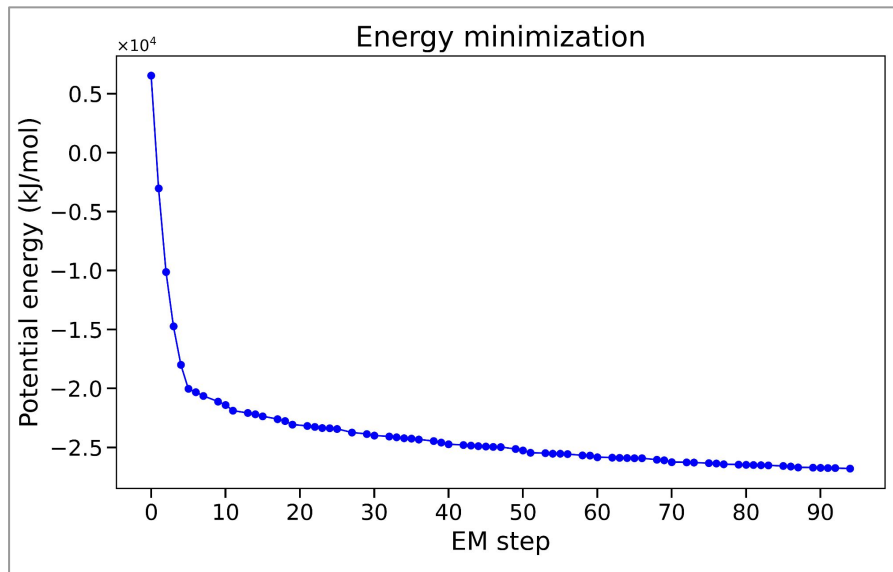
The first thing we need to do is prepare a solvated simulation box containing our biomolecule



Step 2: Energy minimization

Before starting the dynamics, we relax the system through a process called **energy minimization**

- **Why:** remove steric clashes and locally unfavorable geometries introduced during system preparation.
- **How:** atoms are moved iteratively toward lower potential energy, following the forces acting on them.
- **When do we stop:** when the maximum force falls below a chosen threshold.

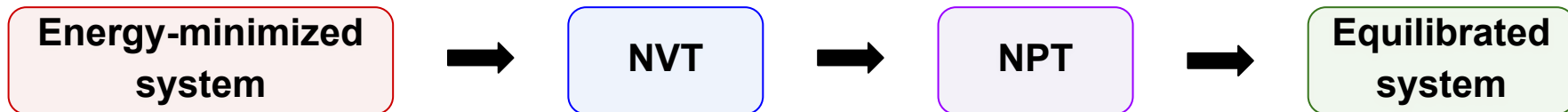


Step 3: Equilibration

After energy minimization, the structure is locally relaxed, but the system is not yet at the desired thermodynamic conditions.

Why do we need to equilibrate?

- Bring the system to the target temperature
- Allow the solvent and ions to reorganize around the peptide
- Adjust the pressure and density
- Avoid starting production MD from an artificial, non-equilibrated state

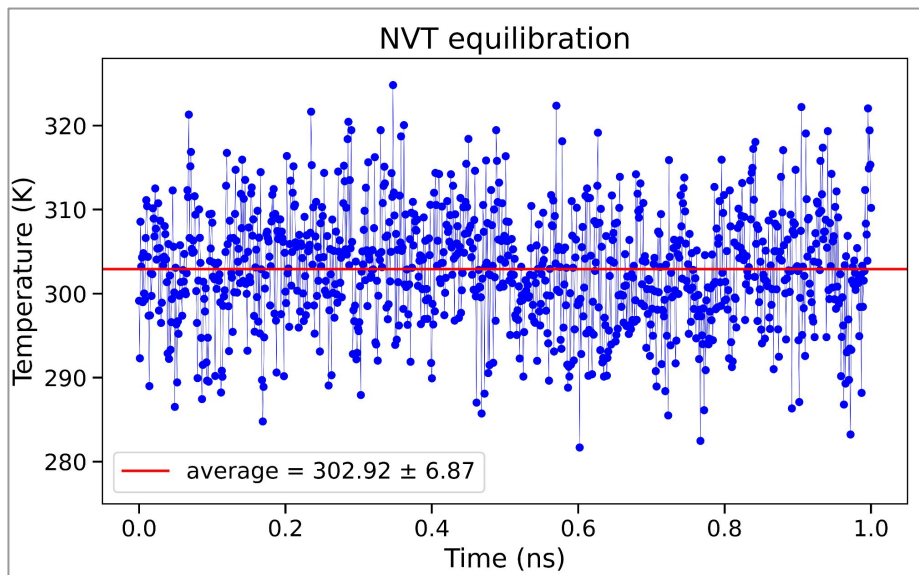


Step 3: NVT equilibration

Goal: bring the system to the target temperature and allow the solvent to reorganize around the peptide

NVT

Number of particles, **V**olume, and **T**emperature are kept **constant**



T is controlled with a thermostat
(stochastic velocity-rescaling)

- T usually equilibrates quickly
- T should fluctuate around the target value with no drift

Step 3: NPT equilibration

Goal: allow the simulation box to adjust until the system reaches the appropriate pressure and density

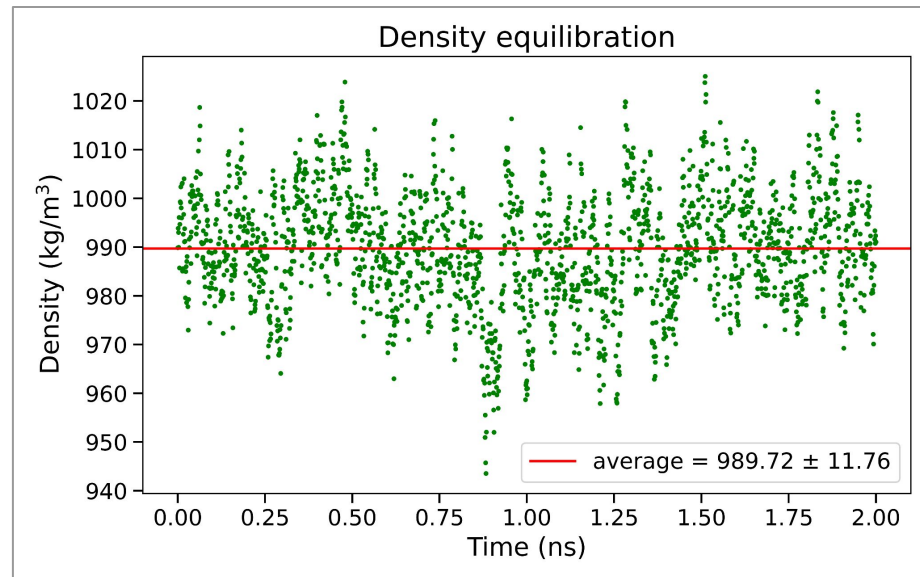
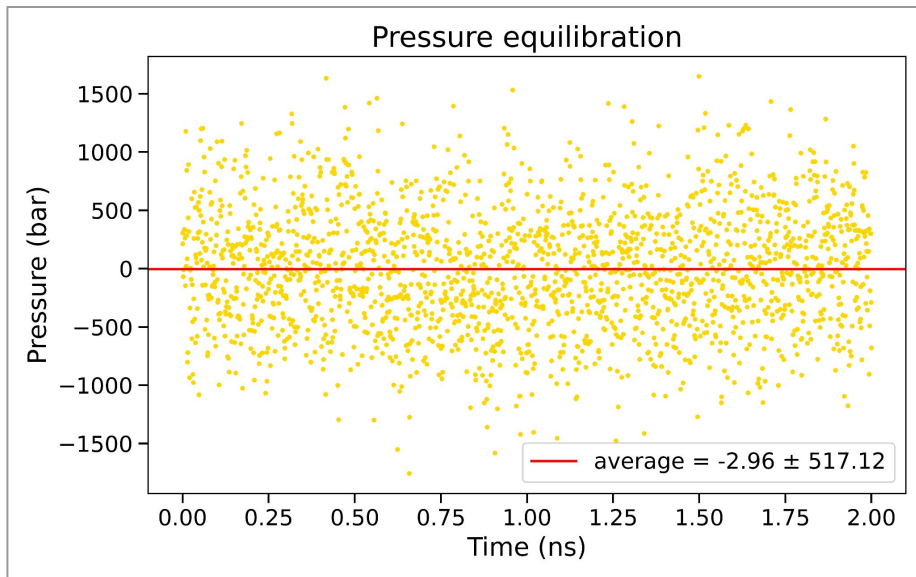
NPT

Number of particles, Pressure, and Temperature are kept **constant**

P is controlled with a barostat
(stochastic cell rescaling)

T is controlled with a thermostat
(stochastic velocity-rescaling)

Step 3: NPT equilibration

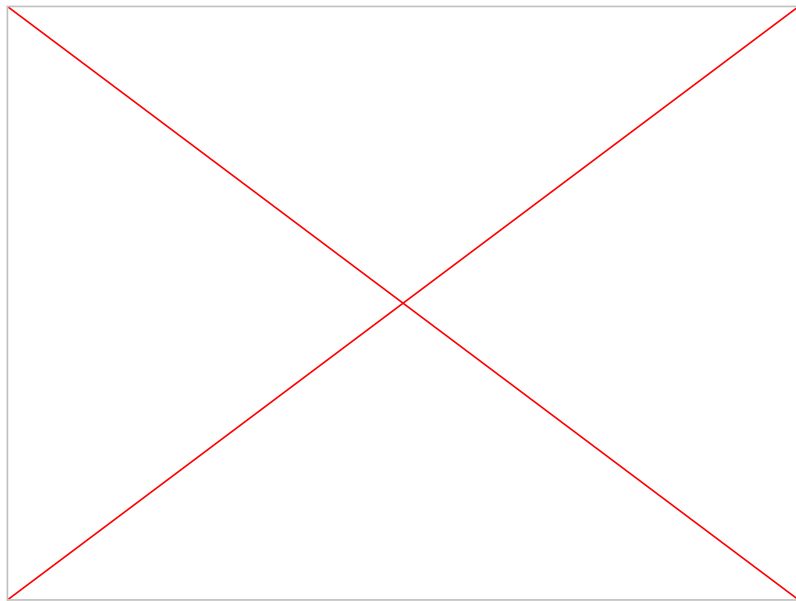


- Instantaneous P fluctuates strongly — this is expected for small MD systems
- Density does not need to match pure water: we are simulating a peptide solution
- We look for a stable average and no systematic drift

Step 4: Production

Once the system is well equilibrated, we can start the production simulation used for data collection.

- **Simulation length** depends on the process of interest (conformational changes, folding/unfolding, ligand binding, etc..)
- Typical atomistic MD scales: ns- μ s
- GPU acceleration now makes longer scales much more accessible
- Coordinates are saved at regular intervals to generate the trajectory used for analysis



Step 5: Analyses

We can perform a variety of analyses on the resulting MD trajectory either using GROMACS itself or python packages such as MDAnalysis.

Structural analysis of the solute

How does the biomolecule itself behave?

- Root mean square deviation (RMSD)
- Root mean square fluctuation (RMSF)
- Radius of gyration (R_g)

Solute-solvent analysis

How does the biomolecule interact with its environment?

- Radial distribution function (RDF)
- Hydrogen bonds
- Solvent accessible surface area (SASA)

Step 5: Solute analysis

Root mean square deviation:

Measures how much the structure at time t differs from a reference structure.

$$\text{RMSD}(t) = \sqrt{\frac{1}{N} \sum_{i=1}^N |r_i(t) - r_i^{ref}|^2}$$

Root mean square fluctuation:

Measures how much each atom i (or residue) fluctuates around its average position.

$$\text{RMSF}_i = \sqrt{\langle |r_i(t) - \langle r_i \rangle|^2 \rangle}$$

Radius of gyration:

Measures how compact (small R_g) or extended (large R_g) the molecule is.

$$R_g = \sqrt{\frac{\sum_i m_i |r_i - r_{\text{COM}}|^2}{\sum_i m_i}}$$

Step 5: Solute-solvent analysis

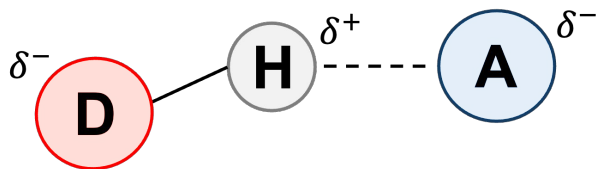
Radial distribution function:

Describes how the local density varies as a function of distance (r) from a reference particle.

$$g(r) = \frac{\rho(r)}{\rho_{\text{bulk}}}$$

Hydrogen bonds:

Describe specific directional interactions between the solute and surrounding water molecules

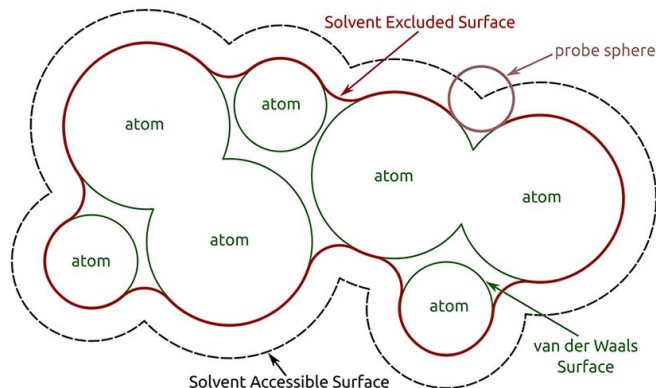


D = donor; **A** = acceptor

$d(\text{D} \cdots \text{A}) \lesssim 3.0 \text{ \AA}$ and $\angle(\text{D-H} \cdots \text{A}) \gtrsim 150^\circ$

Solvent accessible surface area:

Describes the surface area of the solute that is accessible to a solvent



Step 5: Essential dynamics analysis

A trajectory contains thousands of structures — how do we identify the important conformational states?



We need to reduce the dimensionality of this conformational space



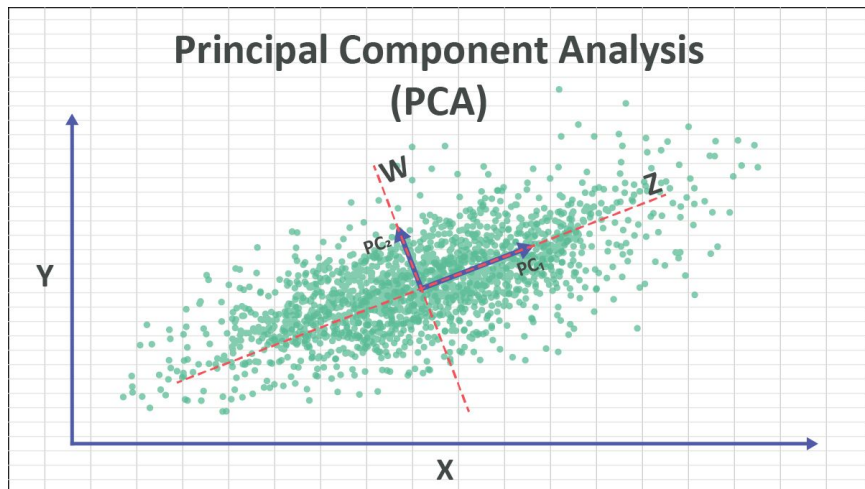
**Essential dynamics
(ED)**

ED applies principal component analysis (PCA) to an MD trajectory to reduce its dimensionality and **reveal the main conformational motions**.

Step 5: Principal component analysis

What is principal component analysis?

PCA is a powerful mathematical technique used in data science, machine learning, and statistics. It transforms high-dimensional data into a **lower-dimensional space** while **preserving** as much of the **variance (spread)** as possible.



Step 5: Principal component analysis

**Set of original
variables**



**Set of uncorrelated
variables (PCs)**



- PCs are linear combinations of the original variables.
- PCs are ordered by the amount of variance they explain

**Dimensionality
reduction**



By keeping only the first few PCs, we reduce the dimensionality of our data while retaining most of its variance.

Step 5: Principal component analysis

Advantages

1. Useful with high-dimensional data
2. It can help filter out noise
3. It makes complex data visualizable
4. It helps identify the directions along which the data varies most

Limitations

1. PCs can be difficult to interpret
2. It only captures linear relationships
3. Some information is inevitably lost when reducing dimensions

What does applying PCA to our MD simulation mean?

Step 5: Essential dynamics analysis

- An MD trajectory of N atoms contains $3N$ Cartesian coordinates per frame ($x_1, y_1, z_1, \dots, x_N, y_N, z_N$)
- Over thousands of frames, this create a **very high-dimensional dataset**
- Atomic motions are often **correlated**: groups of atoms move together rather than independently



- We apply PCA to the **atomic fluctuations** of our system to identify the **collective motions** that dominate its dynamics
- **Goal**: describe the trajectory using only **a few** dominant collective motions instead of all $3N$ coordinates.

Step 5: Essential dynamics analysis

**MD
production**



1) We remove the overall rotation and translation to look at the **internal motion** only (= least square fitting to a ref. structure)

2) We construct the covariance matrix of the positional fluctuations:

$$C = \langle (\mathbf{x} - \langle \mathbf{x} \rangle)(\mathbf{x} - \langle \mathbf{x} \rangle)^T \rangle$$

C is a symmetric $3N \times 3N$ matrix, which can be **diagonalized** with an orthonormal transformation matrix R :

$$R^T C R = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_{3N}), \text{ with } \lambda_1 > \lambda_2 > \dots > \lambda_{3N}$$

Step 5: Essential dynamics analysis

$$\textcircled{R}^T C R = \text{diag}(\lambda_1, \textcircled{\lambda_2}, \dots, \lambda_{3N}), \text{ with } \underline{\lambda_1 > \lambda_2 > \dots > \lambda_{3N}}$$

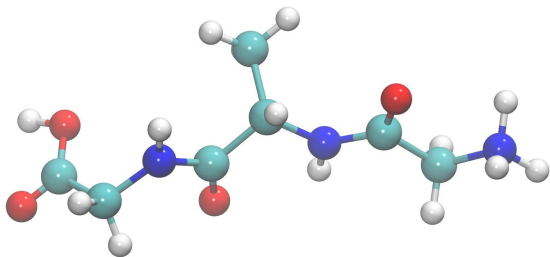
Eigenvectors: columns of R
directions along which the
internal motion takes place;
also called principal/essential
modes.

Eigenvalues:
mean square fluctuations of
the principal components

- The first few principal modes often describe the relevant motion of the system (not a rule)
- The trajectory can be projected on the principal modes to give the **principal components** ($p_i(t)$)

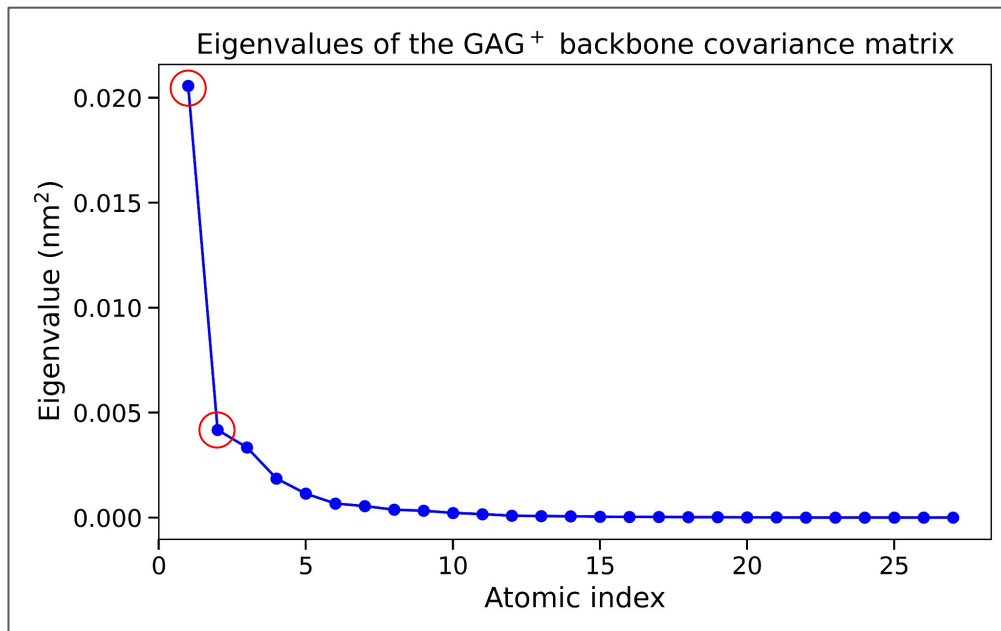
Step 5: Essential dynamics analysis

The first few principal modes often describe the relevant motion of the system



Exercise of this afternoon:
Exploring the conformational landscape
of GAG⁺ in H₂O at T = 30 °C

We focus on the backbone for the ED analysis since it defines the secondary structure (= conformations)



Step 5: Essential dynamics analysis

The trajectory can be projected on the principal modes to give the **principal components**

$$\mathbf{p}(t) = R^T [\mathbf{x}(t) - \langle \mathbf{x} \rangle]$$

Since the first two principal components capture most of the variance, we retain only PC1 and PC2:

$$\mathbf{p}(t) \rightarrow (p_1(t), p_2(t))$$

Each MD frame is therefore represented by one point in a 2D conformational space: $3N \rightarrow 2$

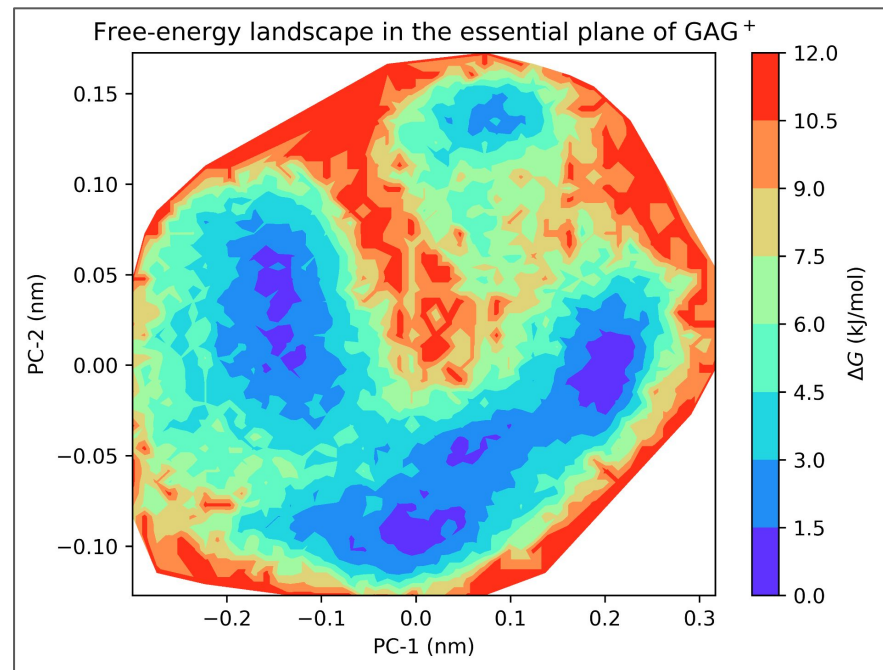
Step 5: Essential dynamics analysis

We build a 2D histogram from the pairs of PC1 and PC2 values sampled along the trajectory.

For each square/bin i , we count how many PC pairs fall inside it $\rightarrow N_i$

$$P_i = \frac{N_i}{N_{max}} = e^{-\Delta G_i/RT}$$

The most populated regions correspond to **free-energy minima** and identify the **preferred conformational states** of the system.



Conclusions

- Classical MD provides an atomistic picture of molecular motion over time
- A reliable simulation requires careful system preparation, energy minimization and equilibration
- The production trajectory can be analyzed to characterize structure, flexibility and solute–solvent interactions
- Essential Dynamics (PCA) reduces the dimensionality of the trajectory and identifies the dominant collective motions
- These approaches work when the relevant states and transitions are sufficiently sampled during the MD simulation

But what if the **transition** between two states occurs on a **timescale much longer** than our MD simulation?